LAPORAN PRAKTIKUM

POSTTEST 5

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

Nathanael Larry Glasia (2509106068)

Kelas (B1‘25)

PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

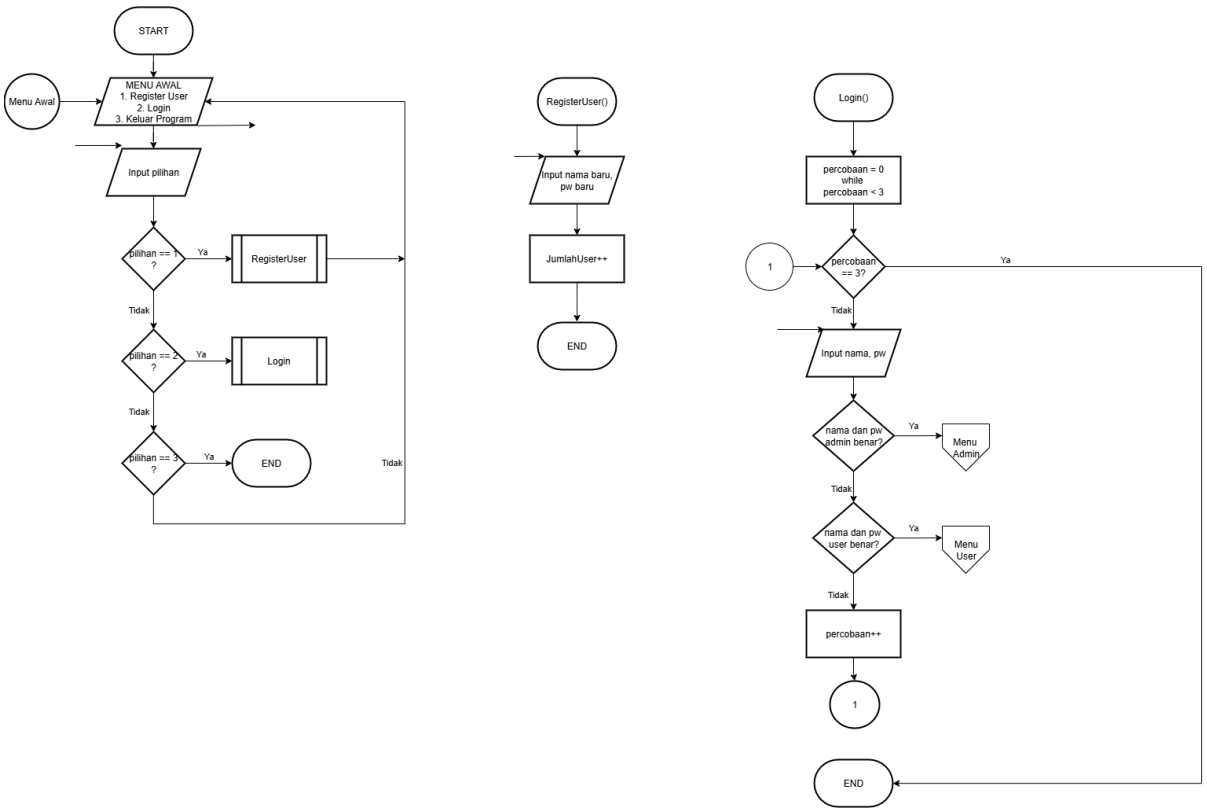
SAMARINDA

2026

1. Flowchart

Program dimulai. Menu awal memiliki 3 pilihan:

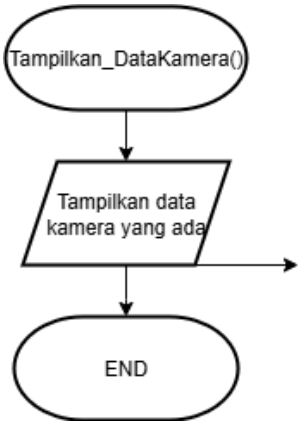
1. Register User
2. Login
3. Keluar Program

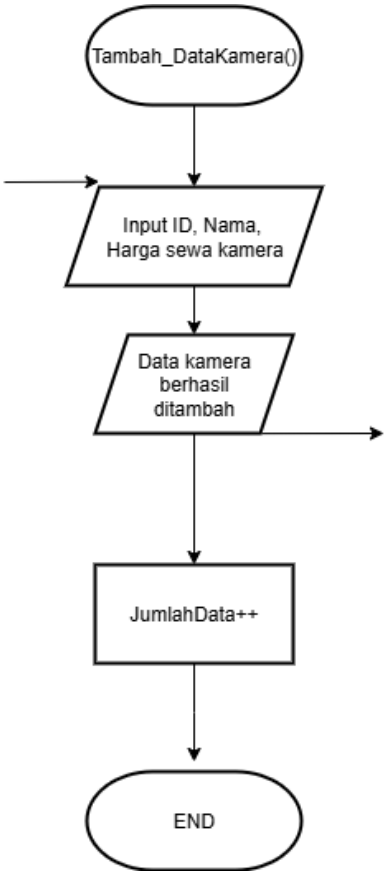


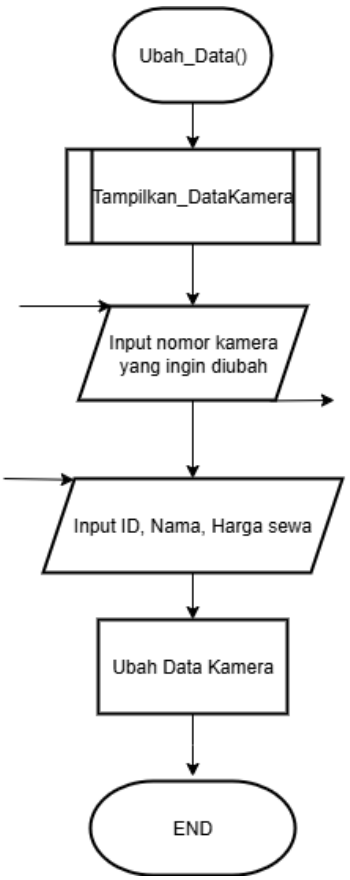
Jika memilih 1, maka pengguna akan diarahkan ke menu register untuk daftar akun baru. Admin diberi nama “LARRY” dan passwordnya “068”, jadi admin tidak perlu register (bisa langsung login ke program). Jika sudah register, pengguna dapat login jika memilih pilihan 2. Di menu login, pengguna juga memasukkan nama dan password yang sesuai ketika pengguna melakukan register akun. Pengguna diberi percobaan login yang diberi nilai 0 dan ada kondisi yang diberikan selama percobaan login kurang dari 3. Jika salah memasukkan nama atau password, maka percobaan login bertambah sebanyak 1 dan jika sampai 3 kali gagal login, maka program otomatis berhenti. Jika memilih 3, maka program akan berhenti.

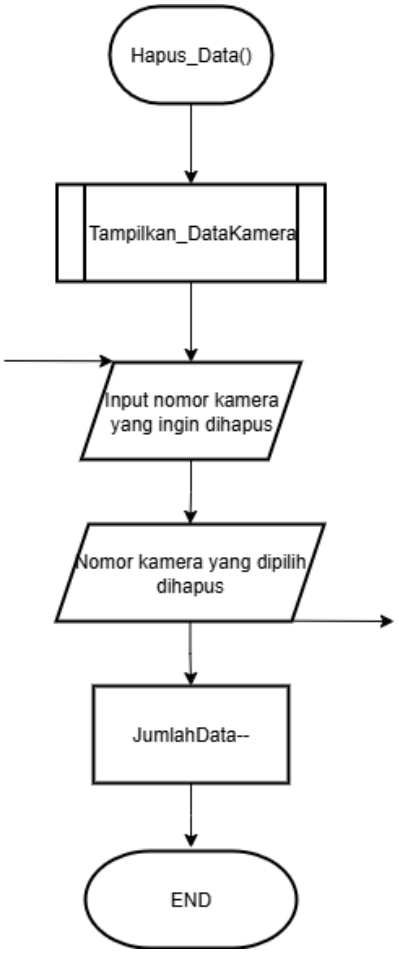
Menu Admin memiliki:

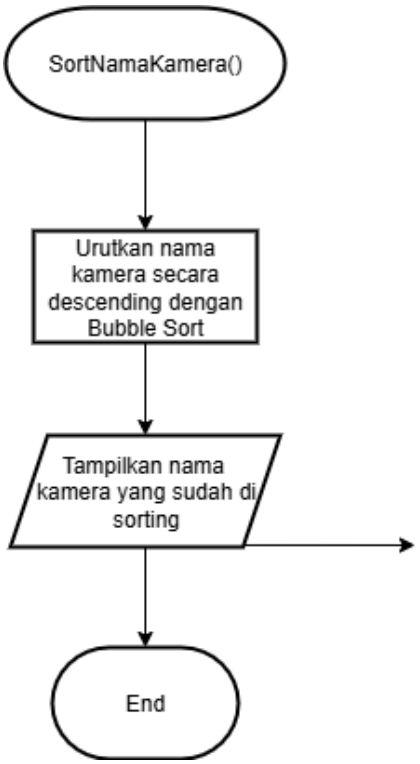
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Sortir Nama Kamera
6. Sortir Harga Sewa Kamera
7. Sortir ID Kamera
8. Keluar ke Menu Awal

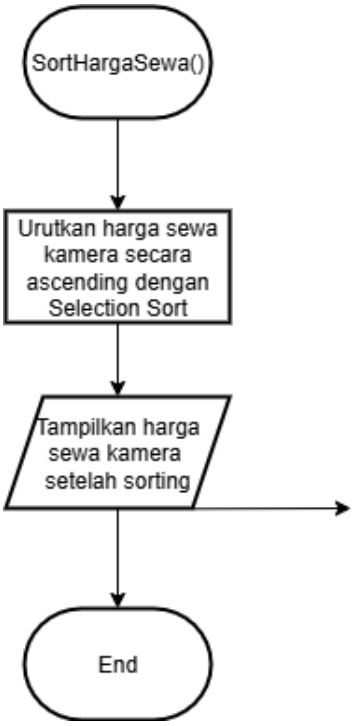


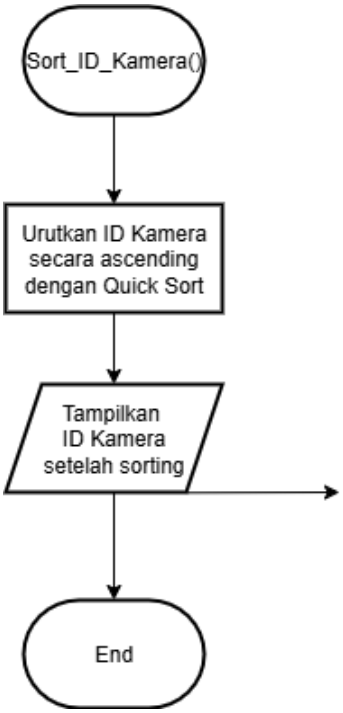

Tambah_DataKamera()
Input ID, Nama,
Harga sewa kamera
Data kamera
berhasil
ditambah
JumlahData++
END


Ubah_Data()
Tampilkan_DataKamera
Input nomor kamera
yang ingin diubah
Input ID, Nama, Harga sewa
Ubah Data Kamera
END

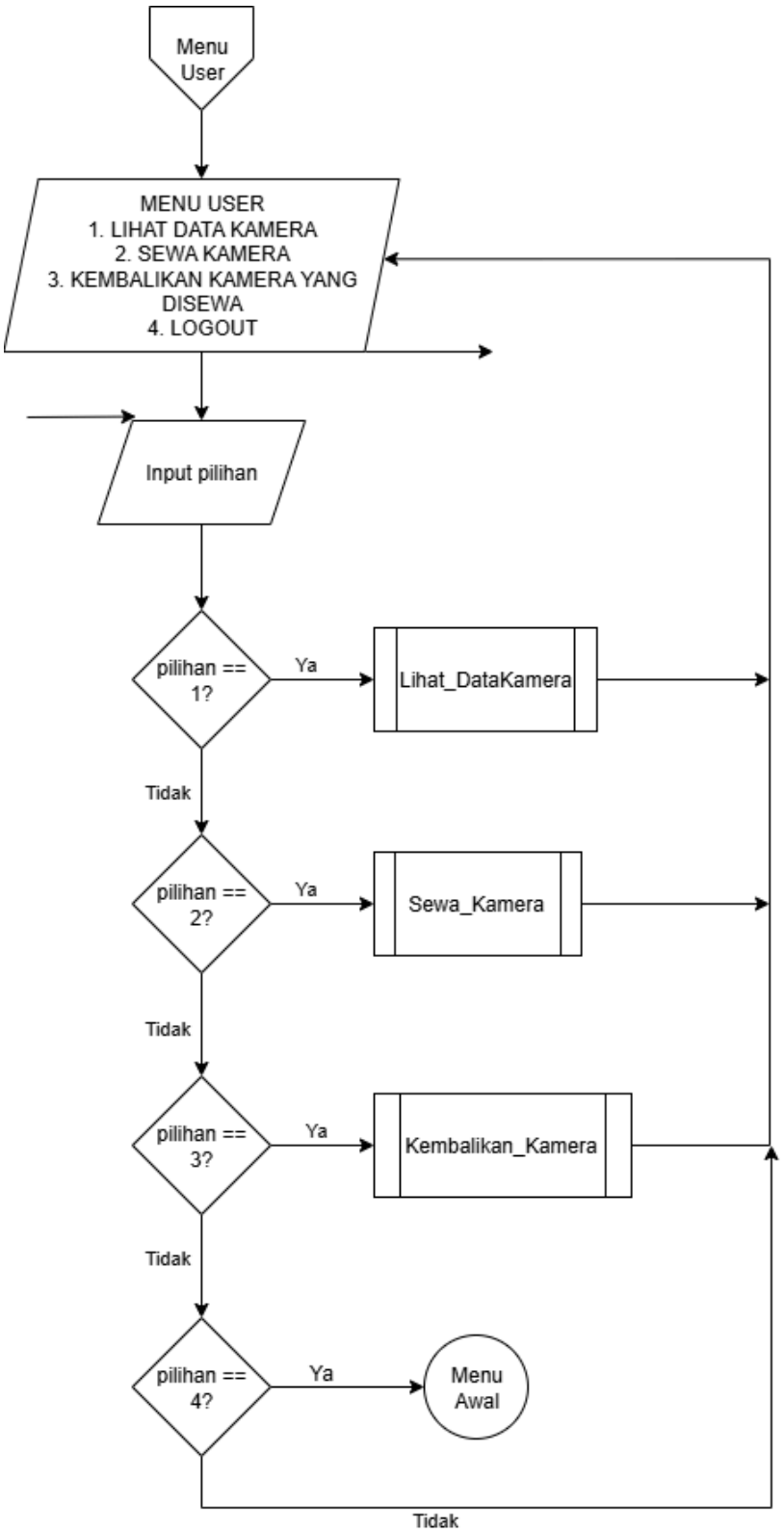

Hapus_Data()
Tampilkan_DataKamera
Input nomor kamera yang ingin dihapus
Nomor kamera yang dipilih dihapus
JumlahData--
END


SortNamaKamera()
Urutkan nama kamera secara descending dengan Bubble Sort
Tampilkan nama kamera yang sudah di sorting
End


SortHargaSewa()
Urutkan harga sewa kamera secara ascending dengan Selection Sort
Tampilkan harga sewa kamera setelah sorting
End

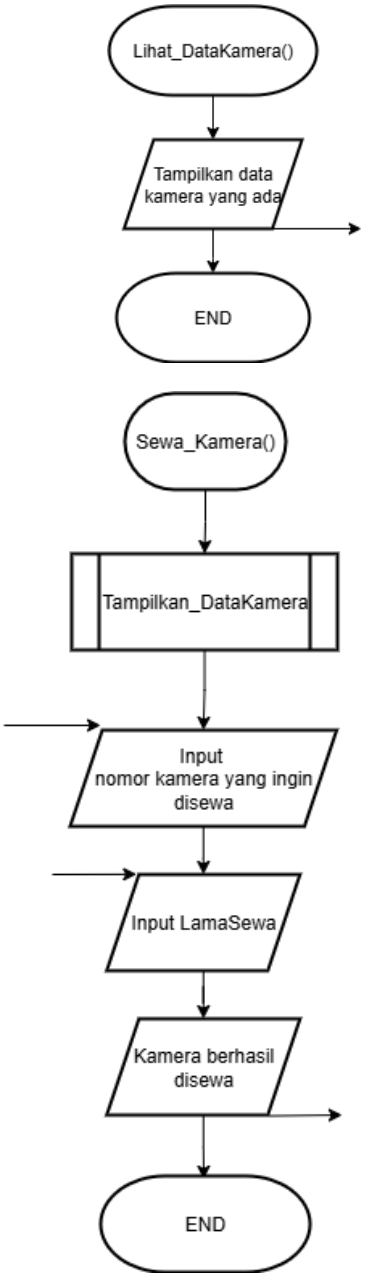

Sort_ID_Kamera()
Urutkan ID Kamera secara ascending dengan Quick Sort
Tampilkan ID Kamera setelah sorting
End

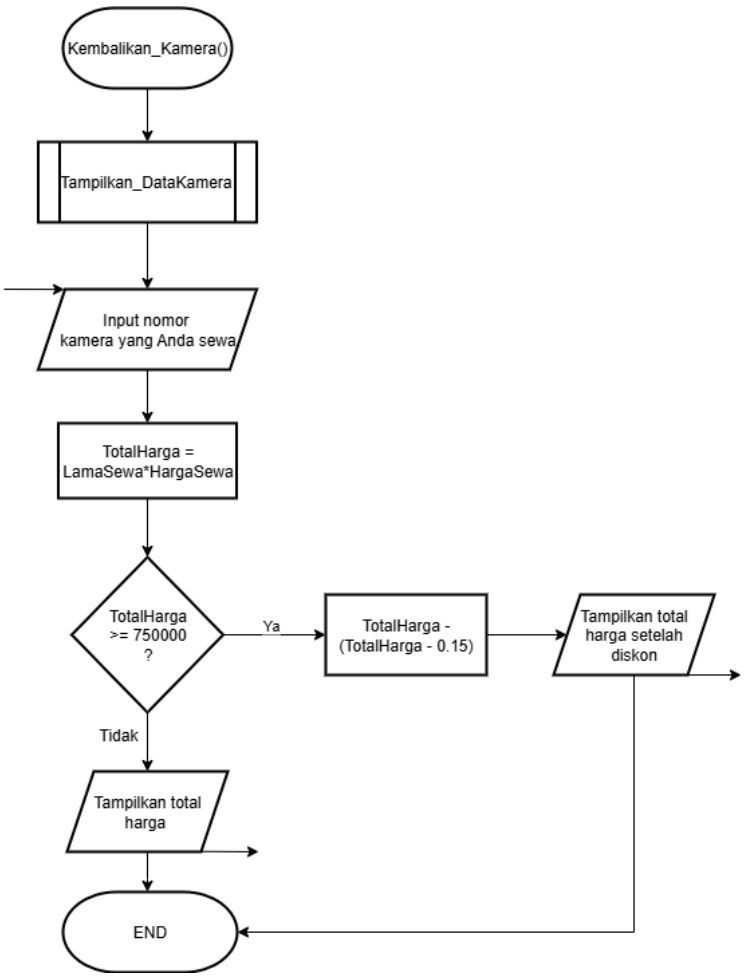
Jika memilih menu nomor 1, maka admin bisa menambah ID kamera, nama kamera, dan harga sewa/hari kamera. Jika memilih menu nomor 2, program akan menampilkan data kamera, baik ID kamera, nama kamera, dan harga sewa per/hari. Jika memilih menu nomor 3, admin dapat mengubah data kamera. Jika memilih menu nomor 4, admin dapat menghapus data kamera. Jika memilih nomor 5, maka admin dapat menyortir nama kamera dengan metode Bubble Sort secara descending. Jika memilih nomor 6, maka admin dapat menyortir harga sewa kamera dengan metode selection sort secara ascending. Jika memilih nomor 7, maka admin dapat menyortir ID Kamera dengan metode Quick Sort secara ascending. Setelah selesai menggunakan menu 1-7, program akan kembali ke menu admin. Jika memilih menu nomor 8, maka program akan mengarahkan ke menu awal.


Menu User
MENU USER
1. LIHAT DATA KAMERA
2. SEWA KAMERA
3. KEMBALIKAN KAMERA YANG DISEWA
4. LOGOUT
Input pilihan
pilihan == 1?
Ya
Lihat_DataKamera
Tidak
pilihan == 2?
Ya
Sewa_Kamera
Tidak
pilihan == 3?
Ya
Kembalikan_Kamera
Tidak
pilihan == 4?
Ya
Menu Awal
Tidak

Menu User memiliki:

1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera yang Disewa
4. Logout





Jika memilih nomor 1, user dapat melihat data kamera yang ada. Jika memilih nomor 2, User dapat menyewa kamera yang tersedia. Jika memilih nomor 3, user dapat mengembalikan kamera yang sebelumnya disewa dan program juga akan menghitung total harga sewa berdasarkan harga sewa dan lama sewa. Setelah selesai menggunakan menu 1-3, program akan kembali ke menu user. Jika memilih nomor 4, maka user kembali ke menu awal.

2. Deskripsi Singkat Program

Program Sistem Data Penyewaan Kamera adalah program yang dimana Admin mengelola dan menyimpan data kamera yang terdiri dari ID kamera, nama kamera, harga sewa/hari, dan status kameranya apakah tersedia atau sedang disewa. User dapat menyewa kamera yang sedang tersedia dan dapat mengembalikan kamera sewaannya sesuai dengan kamera yang disewa dan berapa lama User menyewa kamera itu.

Program menggunakan fungsi dan prosedur untuk memecah beberapa menu dari admin dan user. Pada program ini juga terdapat pointer (* dan &). Dereference Operator (*) adalah operator untuk melihat nilai yang tersimpan pada alamat memori. Sedangkan Address-Of Operator adalah operator untuk mendapatkan atau melihat alamat memori yang dimiliki sebuah variabel. Kedua operator juga bisa digunakan sebagai parameter fungsi ataupun nilai kembalian. Pointer pada struct juga diterapkan untuk mengakses serta memanipulasi data yang ada di dalam struct. Program juga menerapkan sorting untuk mengurutkan data baik secara ascending (terkecil ke terbesar) maupun descending (terbesar ke terkecil).

3. Source Code

```
struct User
{
    string username;
    string password;
    string role;
};

struct StatusKamera
{
    string kondisi;
};

struct Kamera
{
    string ID_Kamera;
    string NamaKamera;
    int HargaSewa;
    int LamaSewa;
    string NamaPenyewa;
    StatusKamera status;
};

void RegisterUser(User *user, int &JumlahUser){
    string username, Password;
    if (JumlahUser >= 2){
        cout << "User penuh";
        return;
    }
    cout << "\n========================\n";
    cout << "      REGISTER USER\n";
    cout << "========================\n";
    cout << "Masukkan nama baru: ";
    getline(cin, (user + JumlahUser)->username);
    cout << "Masukkan password baru: ";
    getline(cin, (user + JumlahUser)->password);
    (user + JumlahUser)-> role = "user";
    JumlahUser++;
    cout << "Registrasi berhasil! Silakan login\n";
}

bool LoginUser(User *user, int &JumlahUser, string &InputNama, string
&role_login){
```

```
    string InputPassword;
    int percobaan_login = 0;

    while (percobaan_login < 3){
        cout << "\n===================\n";
        cout << "        LOGIN\n";
        cout << "===================\n";
        cout << "Masukkan username: ";
        getline(cin, InputNama);
        cout << "Masukkan password: ";
        getline(cin, InputPassword);

        for (int i = 0; i < JumlahUser; i++){
            if (InputNama == (user + i)->username && InputPassword ==
(user + i)->password){
                cout << "Berhasil login! Selamat datang " << InputNama <<
endl;
                role_login = (user + i)->role;
                return true;
            }
        }
        percobaan_login++;
        cout << "Gagal login! Sisa percobaan: " << (3 - percobaan_login)
<< endl;
    }
    cout << "Gagal 3x! Akses ditolak!\n";
    return false;
}

void TambahDataKamera(Kamera *data, int &JumlahData){
    if (JumlahData == 10){
        cout << "Data kamera penuh!";
        return;
    }
    cout <<
"\n==========================================================================
==================" << endl;
    cout << "                          TAMBAH DATA KAMERA TWISTERS CAM
" << endl;
    cout <<
"==========================================================================
================" << endl;
    cout << "ID Kamera: ";
    getline(cin, (data + JumlahData)->ID_Kamera);
```

```
    cout << "Nama Kamera: ";
    getline(cin, (data + JumlahData)->NamaKamera);
    cout << "Harga sewa/hari: Rp";
    cin >> (data + JumlahData)->HargaSewa;
    cin.ignore();
    (data + JumlahData)->status.kondisi = "Tersedia";
    JumlahData++;
    cout << "Data berhasil ditambahkan";
    cout << endl;
}

void TampilkanDataKamera(Kamera *data, int &JumlahData){
    if (JumlahData == 0){
        cout << "Data kamera belum tersedia";
        return;
    }
    cout <<
"\n==========================================================================
==================\n";
    cout << "                                DATA KAMERA TWISTERS CAM\n";
    cout <<
"==========================================================================
================\n";
    cout << left << setw(5) << "No";
    cout << setw(10) << "ID";
    cout << setw(20) << "Nama Kamera";
    cout << setw(20) << "Harga Sewa/hari";
    cout << setw(12) << "Status";
    cout << endl;
    cout <<
"--------------------------------------------------------------------------
---------------\n";

    for (int i = 0; i < JumlahData; i++){
        cout << left << setw(5) << i + 1;
        cout << setw(10) << (data + i) -> ID_Kamera;
        cout << setw(20) << (data + i) -> NamaKamera;
        cout << "Rp" << setw(17) << (data + i) -> HargaSewa;
        cout << setw(12) << (data + i) -> status.kondisi << endl;
    }
    cout <<
"==========================================================================
================\n";
}
```

```
void UbahDataKamera(Kamera *data, int &JumlahData){
    int index;
    if (JumlahData == 0){
        cout << "Data kamera belum tersedia";
        return;
    }
    TampilkanDataKamera(data, JumlahData);
    cout << "Pilih nomor kamera yang ingin diubah: ";
    cin >> index;
    cin.ignore();
    index--;

    if (index >= 0 && index < JumlahData)
    {
        cout << "ID Kamera baru: ";
        getline(cin, (data + index) -> ID_Kamera);
        cout << "Nama Kamera baru: ";
        getline(cin, (data + index) -> NamaKamera);
        cout << "Harga sewa/hari: Rp";
        cin >> (data + index) -> HargaSewa;
        cin.ignore();
        cout << "Status (Tersedia/Disewa): ";
        getline(cin, (data + index) -> status.kondisi);
        cout << "Data kamera berhasil diubah!\n";
        TampilkanDataKamera(data, JumlahData);
    }
    else
    {
        cout << "Nomor tidak valid!";
    }

}

void HapusDataKamera(Kamera *data, int &JumlahData){
    if (JumlahData == 0){
        cout << "Data kamera tidak tersedia!";
        return;
    }
    int index;
    char konfir_hapus;
    TampilkanDataKamera(data, JumlahData);
    cout << "Pilih nomor kamera yang ingin dihapus: ";
    cin >> index;
    cin.ignore();
```

```
    index--;

    if (index >= 0 && index < JumlahData ){
        cout << "Yakin ingin hapus data kamera ini? (Y/N): ";
        cin >> konfir_hapus;

        if (konfir_hapus == 'Y' || konfir_hapus == 'y')
        {
            for (int i = index; i < JumlahData - 1; i++)
            {
                *(data + i) = *(data + i + 1);
            }
            (JumlahData)--;
                cout << "Data kamera berhasil dihapus!\n";
                TampilkanDataKamera(data, JumlahData);
        }
        else
        {
            cout << "Penghapusan dibatalkan!\n";
        }
    }
    else
    {
        cout << "Nomor tidak valid!\n";
    }
}

int KembalikanKamera(int &harga, int &hari){
    if (hari == 0){
        return 0;
    }
    int sisaHari = hari - 1;
    return harga + KembalikanKamera(harga, sisaHari);
}

int KembalikanKamera(int harga, int lamaHari, float diskonSewa){
    return (harga * lamaHari) - ((harga * lamaHari) * diskonSewa);
}

void BubbleSortKameraDescending(Kamera *data, int &JumlahData){
    for (int i = 0; i < JumlahData - 1; i++){
        for (int j = 0; j < JumlahData - i - 1; j++){
            if ((data + j)-> NamaKamera < (data + j + 1)-> NamaKamera){
                Kamera temp = *(data + j);
```

```
                *(data + j) = *(data + j + 1);
                *(data + j + 1) = temp;
            }
        }
    }
}

void SelectionSortHargaAscending(Kamera *data, int &JumlahData){
    for (int i = 0; i < JumlahData - 1; i++){
        int IndexMin = i;
        for (int j = i + 1; j < JumlahData; j++){
            if ((data + j)->HargaSewa < (data + IndexMin)-> HargaSewa){
                IndexMin = j;
            }
        }
        Kamera temp = *(data + i);
        *(data + i) = *(data + IndexMin);
        *(data + IndexMin) = temp;
    }
}

int PartitionID(Kamera *data, int low, int high){
    string pivot = (data + high)-> ID_Kamera;
    int i = low - 1;

    for (int j = low; j < high; j++){
        if ((data + j)-> ID_Kamera < pivot){
            i++;
            Kamera temp = *(data + i);
            *(data + i) = *(data + j);
            *(data + j) = temp;
        }
    }

    Kamera temp = *(data + i + 1);
    *(data + i + 1) = *(data + high);
    *(data + high) = temp;
    return i + 1;
}

void QuickSortID(Kamera *data, int low, int high){
    if (low < high){
        int pivot = PartitionID(data, low, high);
```

```
        int kiriLow = low;
        int kiriHigh = pivot - 1;
        int kananLow = pivot + 1;
        int kananHigh = high;

        QuickSortID(data, low, pivot - 1);
        QuickSortID(data, pivot + 1, high);

    }
}

int main(){
    Kamera data[10];
    User user[2];
    int JumlahUser = 0, JumlahData = 0;
    user[JumlahUser++] = {"LARRY", "068", "admin"};
    bool MenuAwal = true;

    while(MenuAwal)
    {
        int PilihMenu;
        string InputNama, role_login;

        cout << "\n==========================\n";
        cout << "  HOME PAGE TWISTERS CAM\n";
        cout << "==========================\n";
        cout << "1. Register\n";
        cout << "2. Login\n";
        cout << "3. Keluar\n";
        cout << "Pilih (1-3): ";
        cin >> PilihMenu;
        cin.ignore();

        switch(PilihMenu)
        {
            case 1:
            RegisterUser(user, JumlahUser);
            break;

            case 2:
            if(!LoginUser(user, JumlahUser, InputNama, role_login)){
                return 0;
            }
            if (role_login == "admin")
```

```
        {
            int pilihan;
            do
            {
                cout << "\n======================\n";
                cout << "        MENU ADMIN\n";
                cout << "======================\n";
                cout << "1. Tambah Data Kamera\n";
                cout << "2. Tampilkan Data Kamera\n";
                cout << "3. Ubah Data Kamera\n";
                cout << "4. Hapus Data Kamera\n";
                cout << "5. Sortir Nama Kamera\n";
                cout << "6. Sortir Harga Sewa Kamera\n";
                cout << "7. Sortir ID Kamera\n";
                cout << "8. Logout\n";
                cout << "Pilih (1-8): ";
                cin >> pilihan;
                cin.ignore();

                switch(pilihan)
                {
                    case 1:
                    TambahDataKamera(data, JumlahData);
                    break;

                    case 2:
                    TampilkanDataKamera(data, JumlahData);
                    break;

                    case 3:
                    UbahDataKamera(data, JumlahData);
                    break;

                    case 4:
                    HapusDataKamera(data, JumlahData);
                    break;

                    case 5:
                    BubbleSortKameraDescending(data, JumlahData);
                    cout << "Sort Nama Kamera berhasil\n";
                    TampilkanDataKamera(data, JumlahData);
                    break;
```

```
                    case 6:
                    SelectionSortHargaAscending(data, JumlahData);
                    cout << "Sort Harga Kamera berhasil\n";
                    TampilkanDataKamera(data, JumlahData);
                    break;

                    case 7:
                    QuickSortID(data, 0, JumlahData - 1);
                    cout << "Sort ID Kamera berhasil\n";
                    TampilkanDataKamera(data, JumlahData);
                    break;

                    case 8:
                    break;
                }

            } while (pilihan != 8);
        }
        else
        {
            int pilihan;
            do
            {
                cout << "\n======================\n";
                cout << "        MENU USER\n";
                cout << "======================\n";
                cout << "1. Lihat Data Kamera\n";
                cout << "2. Sewa Kamera\n";
                cout << "3. Kembalikan Kamera\n";
                cout << "4. Logout\n";
                cout << "Pilih (1-4): ";
                cin >> pilihan;

                switch (pilihan)
                {
                    case 1:
                    TampilkanDataKamera(data, JumlahData);
                    break;

                    case 2:
                    {
                    if (JumlahData == 0){
                        cout << "Data kamera belum ada";
                        break;
```

```
                        }
                        }

                        {
                            TampilkanDataKamera(data, JumlahData);
                            int pilih;
                            cout << "Pilih nomor kamera yang ingin
disewa: ";
                            cin >> pilih;
                            pilih--;

                            if ((data + pilih)->status.kondisi ==
"Tersedia")
                            {
                                cout << "Sewa berapa hari: ";
                                cin >> (data + pilih)->LamaSewa;
                                (data + pilih)->status.kondisi =
"Disewa";
                                (data + pilih)->NamaPenyewa = InputNama;

                                cout << "Kamera berhasil disewa!\n";
                            }
                            else
                            {
                                cout << "Kamera sedang disewa!\n";
                            }
                            break;
                        }

                        case 3:
                        {
                            if (JumlahData == 0){
                                cout << "Data kamera belum ada";
                                break;
                            }
                        }

                        {
                            TampilkanDataKamera(data, JumlahData);
                            int nomor;
                            cout << "Pilih nomor kamera yang Anda sewa
untuk dikembalikan: ";
                            cin >> nomor;
                            nomor--;
```

```
                                if (nomor < 0 || nomor >= JumlahData){
                                    cout << "Nomor tidak valid\n";
                                    break;
                                }

                                if ((data + nomor)-> NamaPenyewa !=
InputNama){
                                    cout << "Anda tidak menyewa kamera
ini\n";
                                    break;
                                }

                                int total = KembalikanKamera((data +
nomor)->HargaSewa, (data + nomor)->LamaSewa);

                                if (total >= 750000)
                                {
                                    total = KembalikanKamera((data + nomor)->
HargaSewa, (data + nomor)-> LamaSewa, 0.15);
                                }

                                cout << "Total harga sewa yang harus dibayar:
Rp" << total << endl;
                                (data + nomor)->status.kondisi = "Tersedia";
                                (data + nomor)->NamaPenyewa = "";
                                (data + nomor)->LamaSewa = 0;
                                cout << "Kamera berhasil dikembalikan\n";
                                break;
                            }
                        }
                    } while (pilihan != 4);
                }
                break;

            case 3:
            cout << "Program selesai\n";
            return 0;

            default:
            cout << "Pilihan tidak valid\n";
            }
        }
        return 0;
```

```
}
```

4. Hasil Output

```
=======================
        MENU ADMIN
=======================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Sortir Nama Kamera
6. Sortir Harga Sewa Kamera
7. Sortir ID Kamera
8. Logout
Pilih (1-8): 1
```

```
=============================================================================================
                                DATA KAMERA TWISTERS CAM
=============================================================================================
No    ID        Nama Kamera          Harga Sewa/hari      Status
---------------------------------------------------------------------------------------------
1     K01       NIKON A1             Rp234000             Tersedia
2     K03       ZEUS 777             Rp277000             Tersedia
3     K05       NEXGEN A6            Rp150000             Tersedia
4     K07       CONAN ZOOM           Rp575000             Tersedia
5     K09       NIKON 1974           Rp250000             Tersedia
6     K02       SONY 5X              Rp450000             Tersedia
7     K04       DIGICAM              Rp125000             Tersedia
8     K06       SONY A67             Rp450000             Tersedia
9     K08       CANON DX             Rp500000             Tersedia
10    K10       SONIC ALPHA          Rp675000             Tersedia
---------------------------------------------------------------------------------------------
```

```
==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No   ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1    K01       NIKON A1            Rp234000            Tersedia
2    K03       ZEUS 777            Rp277000            Tersedia
3    K05       NEXGEN A6           Rp150000            Tersedia
4    K07       CONAN ZOOM          Rp575000            Tersedia
5    K09       NIKON 1974          Rp250000            Tersedia
6    K02       SONY 5X             Rp450000            Tersedia
7    K04       DIGICAM             Rp125000            Tersedia
8    K06       SONY A67            Rp450000            Tersedia
9    K08       CANON DX            Rp500000            Tersedia
10   K10       SONIC ALPHA         Rp675000            Tersedia
==========================================================================================
Pilih nomor kamera yang ingin diubah: 1
ID Kamera baru: K01
Nama Kamera baru: NEXUS 177
Harga sewa/hari: Rp256000
Status (Tersedia/Disewa): Tersedia
Data kamera berhasil diubah!

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No   ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1    K01       NEXUS 177           Rp256000            Tersedia
2    K03       ZEUS 777            Rp277000            Tersedia
3    K05       NEXGEN A6           Rp150000            Tersedia
4    K07       CONAN ZOOM          Rp575000            Tersedia
5    K09       NIKON 1974          Rp250000            Tersedia
6    K02       SONY 5X             Rp450000            Tersedia
7    K04       DIGICAM             Rp125000            Tersedia
8    K06       SONY A67            Rp450000            Tersedia
9    K08       CANON DX            Rp500000            Tersedia
10   K10       SONIC ALPHA         Rp675000            Tersedia
==========================================================================================
```

```
==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera          Harga Sewa/hari      Status
------------------------------------------------------------------------------------------
1     K01       NEXUS 177            Rp256000            Tersedia
2     K03       ZEUS 777             Rp277000            Tersedia
3     K05       NEXGEN A6            Rp150000            Tersedia
4     K07       CONAN ZOOM           Rp575000            Tersedia
5     K09       NIKON 1974           Rp250000            Tersedia
6     K02       SONY 5X              Rp450000            Tersedia
7     K04       DIGICAM              Rp125000            Tersedia
8     K06       SONY A67             Rp450000            Tersedia
9     K08       CANON DX             Rp500000            Tersedia
10    K10       SONIC ALPHA          Rp675000            Tersedia
==========================================================================================
Pilih nomor kamera yang ingin dihapus: 10
Yakin ingin hapus data kamera ini? (Y/N): y
Data kamera berhasil dihapus!

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera          Harga Sewa/hari      Status
------------------------------------------------------------------------------------------
1     K01       NEXUS 177            Rp256000            Tersedia
2     K03       ZEUS 777             Rp277000            Tersedia
3     K05       NEXGEN A6            Rp150000            Tersedia
4     K07       CONAN ZOOM           Rp575000            Tersedia
5     K09       NIKON 1974           Rp250000            Tersedia
6     K02       SONY 5X              Rp450000            Tersedia
7     K04       DIGICAM              Rp125000            Tersedia
8     K06       SONY A67             Rp450000            Tersedia
9     K08       CANON DX             Rp500000            Tersedia
==========================================================================================
```

```
==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera          Harga Sewa/hari      Status
------------------------------------------------------------------------------------------
1     K03       ZEUS 777             Rp277000            Tersedia
2     K06       SONY A67             Rp450000            Tersedia
3     K02       SONY 5X              Rp450000            Tersedia
4     K09       NIKON 1974           Rp250000            Tersedia
5     K01       NEXUS 177            Rp256000            Tersedia
6     K05       NEXGEN A6            Rp150000            Tersedia
7     K04       DIGICAM              Rp125000            Tersedia
8     K07       CONAN ZOOM           Rp575000            Tersedia
9     K08       CANON DX             Rp500000            Tersedia
==========================================================================================
```

Sort nama kamera secara descending

DATA KAMERA TWISTERS CAM

No	ID	Nama Kamera	Harga Sewa/hari	Status
1	K04	DIGICAM	Rp125000	Tersedia
2	K05	NEXGEN A6	Rp150000	Tersedia
3	K09	NIKON 1974	Rp250000	Tersedia
4	K01	NEXUS 177	Rp256000	Tersedia
5	K03	ZEUS 777	Rp277000	Tersedia
6	K06	SONY A67	Rp450000	Tersedia
7	K02	SONY 5X	Rp450000	Tersedia
8	K08	CANON DX	Rp500000	Tersedia
9	K07	CONAN ZOOM	Rp575000	Tersedia

Sort harga sewa kamera secara ascending

DATA KAMERA TWISTERS CAM

No	ID	Nama Kamera	Harga Sewa/hari	Status
1	K01	NEXUS 177	Rp256000	Tersedia
2	K02	SONY 5X	Rp450000	Tersedia
3	K03	ZEUS 777	Rp277000	Tersedia
4	K04	DIGICAM	Rp125000	Tersedia
5	K05	NEXGEN A6	Rp150000	Tersedia
6	K06	SONY A67	Rp450000	Tersedia
7	K07	CONAN ZOOM	Rp575000	Tersedia
8	K08	CANON DX	Rp500000	Tersedia
9	K09	NIKON 1974	Rp250000	Tersedia

Sort ID Kamera secara ascending

```
==========================
  HOME PAGE TWISTERS CAM
==========================
1. Register
2. Login
3. Keluar
Pilih (1-3): 1

========================
      REGISTER USER
========================
Masukkan nama baru: Bang Satria
Masukkan password baru: 000
Registrasi berhasil! Silakan login

==========================
  HOME PAGE TWISTERS CAM
==========================
1. Register
2. Login
3. Keluar
Pilih (1-3): 2

====================
       LOGIN
====================
Masukkan username: Bang Satria
Masukkan password: 000
Berhasil login! Selamat datang Bang Satria

======================
       MENU USER
======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4):
```

```
==========================================================================================
                          TAMBAH DATA KAMERA TWISTERS CAM
==========================================================================================
ID Kamera: K10
Nama Kamera: CANON 1500D
Harga sewa/hari: Rp600000
Data berhasil ditambahkan

======================
       MENU ADMIN
======================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Logout
Pilih (1-5): 2

==========================================================================================
                           DATA KAMERA TWISTERS CAM
==========================================================================================
No   ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1    K01       NIKON 2000          Rp120000           Tersedia
2    K02       SONY A6000          Rp300000           Tersedia
3    K03       SONY A7             Rp345000           Tersedia
4    K04       SONY A6700          Rp350000           Tersedia
5    K05       CANON 600D          Rp275000           Tersedia
6    K06       CANON 700D          Rp325000           Tersedia
7    K07       CANON R7            Rp400000           Tersedia
8    K08       CANON R100          Rp500000           Tersedia
9    K09       CANON 1200D         Rp500000           Tersedia
10   K10       CANON 1500D         Rp600000           Tersedia
==========================================================================================
```

```
==========================================================================================
                          TAMBAH DATA KAMERA TWISTERS CAM
==========================================================================================
ID Kamera: K01
Nama Kamera: NIKON 2000
Harga sewa/hari: Rp120000
Data berhasil ditambahkan

======================
       MENU ADMIN
======================
1. Tambah Data Kamera
2. Tampilkan Data Kamera
3. Ubah Data Kamera
4. Hapus Data Kamera
5. Logout
Pilih (1-5): 1

==========================================================================================
                          TAMBAH DATA KAMERA TWISTERS CAM
==========================================================================================
```

```
=====================
        LOGIN
=====================
Masukkan username: Bang Satria
Masukkan password: 000
Berhasil login! Selamat datang Bang Satria

=======================
        MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4):
```

```
=====================
        MENU USER
=====================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 1

==========================================================================================
                                DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari      Status
------------------------------------------------------------------------------------------
1     K01       NIKON 2000          Rp120000             Tersedia
2     K02       SONY A6000          Rp300000             Tersedia
3     K03       SONY A7             Rp345000             Tersedia
4     K04       SONY A6700          Rp350000             Tersedia
5     K05       CANON 600D          Rp275000             Tersedia
6     K06       CANON 700D          Rp325000             Tersedia
7     K07       CANON R7            Rp400000             Tersedia
8     K08       CANON R100          Rp500000             Tersedia
9     K09       CANON 1200D         Rp500000             Tersedia
==========================================================================================
```

```
=======================
        MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 2

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     K01       NIKON 2000          Rp120000            Tersedia
2     K02       SONY A6000          Rp300000            Tersedia
3     K03       SONY A7             Rp345000            Tersedia
4     K04       SONY A6700          Rp350000            Tersedia
5     K05       CANON 600D          Rp275000            Tersedia
6     K06       CANON 700D          Rp325000            Tersedia
7     K07       CANON R7            Rp400000            Tersedia
8     K08       CANON R100          Rp500000            Tersedia
9     K09       CANON 1200D         Rp500000            Tersedia
==========================================================================================
Pilih nomor kamera yang ingin disewa: 5
Sewa berapa hari: 3
Kamera berhasil disewa!
```

```
=======================
        MENU USER
=======================
1. Lihat Data Kamera
2. Sewa Kamera
3. Kembalikan Kamera
4. Logout
Pilih (1-4): 3

==========================================================================================
                              DATA KAMERA TWISTERS CAM
==========================================================================================
No    ID        Nama Kamera         Harga Sewa/hari     Status
------------------------------------------------------------------------------------------
1     K01       NIKON 2000          Rp120000            Tersedia
2     K02       SONY A6000          Rp300000            Tersedia
3     K03       SONY A7             Rp345000            Tersedia
4     K04       SONY A6700          Rp350000            Tersedia
5     K05       CANON 600D          Rp275000            Disewa
6     K06       CANON 700D          Rp325000            Tersedia
7     K07       CANON R7            Rp400000            Tersedia
8     K08       CANON R100          Rp500000            Tersedia
9     K09       CANON 1200D         Rp500000            Tersedia
==========================================================================================
Pilih nomor kamera yang Anda sewa untuk dikembalikan: 5
Diskon: Rp123750
Harga sewa sebelum diskon: Rp825000
Total harga sewa yang harus dibayar: Rp701250
Kamera berhasil dikembalikan
```

5. Langkah-langkah GIT

5.1 GIT Add

```
PS C:\praktikum-apl>  git add 'post-test/post-test-apl-5/2509106068-NATHANAEL LARRY GLASIA-PT-5.cpp'
```

Menambahkan file ke staging area untuk di commit

5.2 GIT Commit

```
PS C:\praktikum-apl> git commit -m "pt 5"
[main 68e6749] pt 5
 1 file changed, 462 insertions(+)
 create mode 100644 post-test/post-test-apl-5/2509106068-NATHANAEL LARRY GLASIA-PT-5.cpp
```

Menyimpan perubahan yang ada di staging area ke repositori lokal dengan pesan

5.3 Git Push

```
PS C:\praktikum-apl> git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 3.09 KiB | 633.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/larrynathanx/praktikum-apl.git
   52c8b93..68e6749  main -> main
```

Mengunggah perubahan dari repositori lokal ke repositori jarak jauh (GitHub)